

2026 1st April

derivative

Let $f(x) = \frac{4x^3+x^2}{x+1}$, **find** $f'(3)$

$$\frac{(12x^2+2x)(x+1)-(4x^3+x^2)(1)}{(x+1)^2} = \frac{12x^3+12x^2+2x^2+2x-4x^3-x^2}{x^2+2x+1} = \frac{8x^3+13x^2+2x}{x^2+2x+1} = \\ = x \frac{8x^2+13x+2}{x^2+2x+1} \Big|_{x=3} = \frac{339}{16}$$

Let $f(x) = e^{\sin(2x)}$, **find** $f'''(0)$

$$f' = e^{\sin(2x)} \cdot 2 \cos(2x)$$

$$f'' = (e^{\sin(2x)} \cdot 2 \cos(2x)) \cdot 2 \cos(2x) + e^{\sin(2x)} \cdot (-2 \sin(2x)) = \\ = 4e^{\sin(2x)} \cos^2(2x) - 4e^{\sin(2x)} \sin(2x) = 4e^{\sin(2x)} (\cos^2(2x) - \sin(2x))$$

$$f''' = 4e^{\sin(2x)} \cdot 2 \cos(2x) (\cos^2(2x) - \sin(2x)) + 4e^{\sin(2x)} [(2 \cos(2x))(-\sin(2x)) - 2 \cos(2x)] \Big|_{x=0} = 0$$

integral

$$\int_{\ln 6}^{\ln 7} \frac{5e^{3u}+e^{2u}+5e^u+1}{\cosh(u)} \, du$$

$$x = e^u, \quad u = \ln x, \quad du = \frac{1}{x}$$

$$\int_{e^{\ln 6}}^{e^{\ln 7}} \frac{5x^3+x^2+5x+1}{\cosh(\ln x)} \cdot \frac{1}{x} \, dx,$$

$$\cosh(u) = \frac{e^u + e^{-u}}{2} \implies \cosh(\ln(x)) = \frac{e^{\ln x} + e^{-\ln x}}{2} = \frac{x + \frac{1}{x}}{2},$$

$$\int_{e^{\ln 6}}^{e^{\ln 7}} \frac{5x^3+x^2+5x+1}{\frac{x^2+1}{2}} \, dx = 2 \int_{e^{\ln 6}}^{e^{\ln 7}} \frac{5x^3+x^2+5x+1}{x^2+1} \, dx = 2 \int_{e^{\ln 6}}^{e^{\ln 7}} \frac{(x^2+1)(5x+1)}{x^2+1} \, dx = 2 \int_{e^{\ln 6}}^{e^{\ln 7}} 5x+1 \, dx = \\ = 2 \left[5 \frac{x^2}{2} + x \right]_{e^{\ln 6}}^{e^{\ln 7}} = 2 \left(\frac{245}{2} + 7 - 90 - 6 \right) = 67$$

2026 2nd April

derivative

Let $f(x) = \sqrt{2x^2 + 1}$, **find** $f'(2)$

$$\sqrt{2x^2 + 1} = (2x^2 + 1)^{\frac{1}{2}} \xrightarrow{0'} \frac{1}{2}(2x^2 + 1)^{-\frac{1}{2}}(4x) = \frac{4x}{2\sqrt{2x^2 + 1}} = \frac{2x}{\sqrt{2x^2 + 1}}$$

$$f'(2) = \frac{2 \cdot 2}{\sqrt{2 \cdot 4 + 1}} = \frac{4}{3}$$

Let $f(x) = e^{2x} \sec(3x)$, **find** $f'(\frac{\pi}{12})$

$$(e^{2x} \sec(3x))' = 2e^{2x} \sec(3x) + e^{2x} 3 \tan(3x) \sec(3x) = e^{2x} (2 \sec(3x) + 3 \tan(3x) \sec(3x)) = e^{2x} (\sec(3x) (2 + 3 \tan(3x)))$$

$$f'(\frac{\pi}{12}) = e^{\frac{\pi}{6}} \left(\sec\left(\frac{\pi}{4}\right) \left(2 + 3 \tan\left(\frac{\pi}{4}\right)\right) \right)$$

integral

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos^{11}(x) + \cos(x)} dx$$

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos^{10}(x) \cos(x) + \cos(x)} dx = \int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos(x)(\cos^{10}(x) + 1)} = \int_0^{\frac{\pi}{4}} \frac{\tan(x)}{\cos^{10}(x) + 1} \quad \text{hell no}$$

$$\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos^{11}(x) + \cos(x)} dx, \quad u = \cos(x), \quad du = -\sin(x) dx$$

$$\int \frac{\sin(x)}{u^{11} + u} (-\sin(x)) du = - \int \frac{1}{u^{11} + u} du = - \left(\int \frac{1}{u^{11}} du + \int \frac{1}{u} du \right) = - \left[\frac{u^{-10}}{-10} + \ln(u) \right],$$

$$\left[\frac{1}{10u^{10}} - \ln(u) \right] \Rightarrow \left[\frac{1}{10 \cos^{10}(x) - \ln(\cos(x))} \right]_0^{\frac{\pi}{4}} \quad \text{dam i messed up bigtime}$$

2026 3rd April

derivative

Let $g(t) = 4e^{t^2-3t+2}$, **find** $g'(1)$

$$\left(4e^{t^2-3t+2}\right)' = 4(2t-3)e^{t^2-3t+2}$$

$$g'(t) = -4$$

Let $f(x) = \frac{e^{x^2-4}}{\cos(\pi x)}$, **find** $f'(2)$

$$\left(\frac{e^{x^2-4}}{\cos(\pi x)}\right)' = \frac{2xe^{x^2-4}\cos(\pi x) + \sin(\pi x)e^{x^2-4}}{\cos^2(\pi x)} = \frac{e^{x^2-4}(2x\cos(\pi x) + \sin(\pi x))}{\cos^2(\pi x)}$$

$$f'(2) = \frac{1(4 \cdot 1 + 0)}{1^2} = 4$$

Let $f(x) = 5e^{\arcsin(\cos(2x))}$, **find** $f'(\frac{\pi}{12})$

$$\left(5e^{\arcsin(\cos(2x))}\right)' = 5e^{\arcsin(\cos(2x))} \left(-\frac{\sin(x)}{\sqrt{1-\cos^2(2x)}}\right)$$

$$f'\left(\frac{\pi}{12}\right) = 5e^{\frac{\pi}{3}} \left(-\frac{\frac{\sqrt{6}-\sqrt{2}}{4}}{\sqrt{1-\frac{3}{4}}}\right)$$

well i guess i fucked up

integral

$$\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \sin(2\theta)(5\cot(\theta) + 3\tan(\theta)) \, d\theta$$

$$\cot = \frac{1}{\tan}, \quad \tan = \frac{\sin}{\cos}, \implies \cot = \frac{\cos}{\sin},$$

$$\sin(2\theta)(5\cot(\theta) + 3\tan(\theta)) = 2\sin(\theta)\cos(\theta) \left(5\frac{\cos(\theta)}{\sin(\theta)} + 3\frac{\sin(\theta)}{\cos(\theta)}\right) = 10\cos^2(\theta) + 6\sin^2(\theta) =$$

$$= 10\frac{1+\cos(2\theta)}{2} + 6\frac{1-\cos(2\theta)}{2},$$

$$5\left(\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} 1 \, d\theta + \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \cos(2\theta) \, d\theta\right) + 3\left(\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} 1 \, d\theta - \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \cos(2\theta) \, d\theta\right) = \frac{2\pi}{3} + 2\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \cos(2\theta) \, d\theta = \frac{2\pi}{3} + 2\left[\frac{\sin(2\theta)}{2}\right]_{\frac{\pi}{12}}^{\frac{\pi}{6}} =$$

$$= \frac{2\pi}{3} + [\sin(2\theta)]_{\frac{\pi}{12}}^{\frac{\pi}{6}} = \frac{2\pi}{3} + \left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right) = \frac{2\pi}{3} + \frac{\sqrt{3}-1}{2}$$

$$\int_0^{\frac{3\pi}{4}} \ln\left(1 + \sqrt{\sin(x) + \cos(x)}\right)(\sin(x) - \cos(x)) \, dx$$

$$u = \sin(x) + \cos(x), \quad \frac{du}{dx} = \cos(x) - \sin(x), \quad dx = \frac{du}{\cos(x) - \sin(x)}$$

$$\int_1^0 \ln(1 + \sqrt{u})(\sin(x) - \cos(x)) \frac{1}{\cos(x) - \sin(x)} \, du = \int_1^0 \ln(1 + \sqrt{u}) \frac{-(\cos(x) - \sin(x))}{\cos(x) - \sin(x)} \, du = \int_1^0 -\ln(1 + \sqrt{u}) \, du,$$

$$u = w^2, \quad \frac{du}{dw} = 2w, \quad du = 2w \, dw,$$

$$\int_1^0 -\ln(1 + \sqrt{w^2}) 2w \, dw = -2 \int_1^0 \ln(1 + w) w \, dw,$$

	D	I
+	$\ln(1 + w)$	w
-	$\frac{1}{1+w}$	$\frac{w^2}{2}$

$$-2 \left(\frac{\ln(1 + w)w^2}{2} - \frac{1}{2} \int_1^0 \frac{w^2}{1 + w} \, dw \right) = -w^2 \ln(1 + w) + \int_1^0 \frac{w^2}{1 + w} \, dw,$$

$$\begin{aligned} \int_1^0 \frac{w^2}{1 + w} \, dw &= \int_1^0 \frac{w^2 - 1 + 1}{1 + w} \, dw = \int_1^0 \left(\frac{w^2 - 1}{w + 1} + \frac{1}{w + 1} \right) \, dw = \int_1^0 \left(\frac{(w - 1)(w + 1)}{w + 1} + \frac{1}{w + 1} \right) \, dw = \\ &= \int_1^0 w \, dw - \int_1^0 1 \, dw + \int_1^0 \frac{1}{w + 1} \, dw \end{aligned}$$

$$\left[-w^2 \ln(1 + w) + \frac{1}{2}w^2 - w + \ln|w + 1| \right]_1^0 = 0 - \left(-\ln(2) + \frac{1}{2} - 1 + \ln(2) \right) = \frac{1}{2}$$

limit

$$\lim_{n \rightarrow \infty} \sum_{r=2n+1}^{3n} \sum_{k=-6n}^{6n} \frac{1}{r+6kn}$$

$$\text{Mittag-Leffler expansion: } \pi \cot(\pi x) = \lim_{N \rightarrow \infty} \sum_{k=-N}^N \frac{1}{x + k}$$

$$\sum_{k=-6n}^{6n} \frac{1}{r + 6kn}, = \frac{1}{6n} \sum_{k=-6n}^{6n} \frac{1}{\frac{r}{6n} + k} \xrightarrow{n \rightarrow \infty} \frac{1}{6n} \pi \cot\left(\pi \frac{r}{6n}\right),$$

$$\lim_{n \rightarrow \infty} \sum_{r=2n+1}^{3n} \frac{1}{6n} \pi \cot\left(\pi \frac{r}{6n}\right) = \int_{\frac{1}{3}}^{\frac{1}{2}} \pi \cot(\pi x) \, dx,$$

$$u = \pi x, \quad \frac{du}{dx} = \pi, \quad dx = \frac{du}{\pi},$$

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \pi \cot(u) \frac{1}{\pi} \, du = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot(u) \, du = [\ln|\sin(u)|]_{\frac{\pi}{3}}^{\frac{\pi}{2}} = -\ln\left(\frac{\sqrt{3}}{2}\right) = \ln\left(\frac{2}{\sqrt{3}}\right)$$

2026 4th April

derivative

Let $f(x) = \arctan\left(\frac{1-x^2}{2x}\right)$, **find** $f'\left(\frac{1}{2}\right)$

$$f'(x) = \frac{1}{1 + \left(\frac{1-x^2}{2x}\right)^2} \underbrace{\left(\frac{-2x2x - (1-x^2)2}{(2x)^2}\right)}_{\frac{-x^2-1}{2x^2}} = \frac{\frac{-x^2-1}{2x^2}}{1 + \left(\frac{1-x^2}{2x}\right)^2}$$

$$f'\left(\frac{1}{2}\right) = \frac{\frac{-\frac{1}{4}-1}{\frac{1}{2}}}{1 + \left(\frac{1-\frac{1}{4}}{\frac{1}{2}}\right)^2} = \frac{-\frac{5}{2}}{\frac{25}{16}} = -\frac{5}{2} \frac{16}{25} = -\frac{80}{50} = -\frac{8}{5}$$

Let $f(x) = \ln(1 + \sin(x) - x)$, **find** $f'(0)$

$$f'(x) = \frac{\cos(x) - 1}{1 + \sin(x) - x}$$

$$f'(0) = \frac{0}{1 + 1 - 0} = 0$$

integral

$$\int_1^{e^\pi} \cos(\ln(x)) \, dx$$

$$u = \ln(x), \quad \frac{du}{dx} = \frac{1}{x} \implies dx = x \, du, \quad x = e^u$$

$$\int_0^\pi \cos(u) e^u \, du,$$

	D	I
+	e^u	$\cos(u)$
-	e^u	$\sin(u)$
+	e^u	$-\cos(u)$

$$e^u \sin(u) + e^u \cos(u) - \int_0^\pi \cos(u) e^u \, du,$$

$$2I = e^u (\sin(u) + \cos(u))$$

$$I = \frac{e^u (\sin(u) + \cos(u))}{2}$$

$$\left[\frac{e^u (\sin(u) + \cos(u))}{2} \right]_0^\pi = -\frac{e^\pi}{2} - \frac{1}{2}$$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (2x + 2x \cot^2(x) + 2x \tan^2(x) + 2x \tan^2(x) \cot^2(x)) \, \mathrm{d}x$$

$$2x + 2x \cot^2(x) + 2x \tan^2(x) + 2x \tan^2(x) \cot^2(x) = 2x(\cot^2(x) + \tan^2(x) + 2) =$$

$$= 2x((1 + \cot^2(x)) + (1 + \tan^2(x))) = 2x(\csc^2(x) + \sec^2(x))$$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 2x(\csc^2(x) + \sec^2(x)) \, \mathrm{d}x$$

	D	I
$+$	$2x$	$\csc^2(x) + \sec^2(x)$
$-$	2	$\tan(x) - \cot(x)$
$+$	0	$-\ln \cos(x) - \ln \sin(x) $

$$[2x(\tan(x) - \cot(x)) - 2(-\ln|\cos(x)| - \ln|\sin(x)|)]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = [2(x(\tan(x) - \cot(x)) + \ln|\cos(x)| + \ln|\sin(x)|)]_{\frac{\pi}{6}}^{\frac{\pi}{4}} =$$

$$= \frac{2\pi\sqrt{3}}{9} + \ln\left(\frac{4}{3}\right)$$

limit

$$\lim_{n\rightarrow\infty}\sum_{m=0}^n\frac{1}{\int_0^{2^m}\{x\}\,\mathrm{d}x}$$

$$\int_0^{2^m}\{x\}\,\mathrm{d}x=2^m\cdot\frac{1}{2}=2^{m-1},$$

$$\lim_{n\rightarrow\infty}\sum_{m=0}^n\frac{1}{2^{m-1}}=\frac{2}{1-\frac{1}{2}}=4$$

2026 5th April

derivative

easy

trivial, $f'(1) = \frac{1}{2\sqrt{2}}$

medium

Let $f(x) = \sqrt{\ln(\sqrt{\ln(\sqrt{x})})}$, find $f'(8)$

$$\begin{aligned}a &= \sqrt{b}, \quad b = \ln(c), \quad c = \sqrt{d}, \quad d = \ln(e), \quad e = \sqrt{x}, \\a' &= \frac{1}{2\sqrt{b}}, \quad b' = \frac{1}{c}, \quad c' = \frac{1}{2\sqrt{d}}, \quad d' = \frac{1}{e}, \quad e' = \frac{1}{2\sqrt{x}}, \\f'(x) &= \frac{1}{2\sqrt{b}} \cdot \frac{1}{c} \cdot \frac{1}{2\sqrt{d}} \cdot \frac{1}{e} \cdot \frac{1}{2\sqrt{x}}, \\f'(8) &= \frac{1}{96 \ln(2) \sqrt{\ln(\sqrt{\frac{3}{2} \ln(2)})}}\end{aligned}$$

integral

beginner

Let $f(x) = \int_4^{12} \sqrt{8x+4} e^{\sqrt{2x+1}} dx$

$$\int_4^{12} \sqrt{8x+4} e^{\sqrt{2x+1}} dx = \int_4^{12} \sqrt{4(2x+1)} e^{\sqrt{2x+1}} dx = \int_4^{12} \sqrt{4} \sqrt{2x+1} e^{\sqrt{2x+1}} dx,$$

$$u = \sqrt{2x+1}, \quad \frac{du}{dx} = \frac{1}{2\sqrt{2x+1}} \cdot 2 \Rightarrow \frac{du}{dx} = \frac{1}{u}, \quad dx = u du,$$

$$\int_3^5 2ue^u u du = 2 \int_3^5 u^2 e^u du$$

	D	I
$+$	u^2	e^u
$-$	$2u$	e^u
$+$	2	e^u
$-$	0	e^u

$$2[e^u(u^2 - 2u + 2)]_3^5 = 2[17e^5 - 5e^3]$$

easy

Let $\int_0^1 x \ln(x^2 + 1) dx$

$$u = x^2 + 1, \quad \frac{du}{dx} = 2x, \quad dx = \frac{du}{2x}, \quad x = \sqrt{u-1},$$

$$\int_1^2 \sqrt{u-1} \ln(u) \frac{1}{2\sqrt{u-1}} du = \frac{1}{2} \int_1^2 \ln(u) du = \frac{1}{2} [u \ln(u) - u]_1^2 = \frac{1}{2} [(2 \ln(2) - 2) + 1] = \ln(2) - \frac{1}{2}$$

limit

Let $f(x) = \frac{\arctan(x)}{x}$, find $\lim_{x \rightarrow \infty} f^{(4)}(x)$

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots$$

$$f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \frac{x^6}{7} + \frac{x^8}{9} - \dots$$

$$f^{(4)}(x) = 0 - 0 + \frac{24}{5} - ? x^2 + ? x^4$$

2026 6th April

derivative

easy

Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3)$, $x \in \mathbb{R}$, find $f'(5)$

$$A = f'(1), \quad B = f''(2), \quad C = f'''(3),$$

$$f(x) = x^3 + Ax^2 + 2Bx + C, \quad f'(x) = 3x^2 + 2xA + 2B, \quad f''(x) = 6x + 2A, \quad f'''(x) = 6,$$

$$f'''(3) = 6 = C,$$

$$f''(2) = 12 + 2A = B,$$

$$f'(1) = 3 + 2A + 2B = A \implies A = 3 + 2A + 2(12 + 2A) \implies A = 27 + 6A \implies A = -\frac{27}{5},$$

$$B = 12 + 2\left(-\frac{27}{5}\right) = \frac{60}{5} - \frac{54}{5} \implies B = \frac{6}{5},$$

$$f'(5) = 75 + 10\left(-\frac{27}{5}\right) + \frac{12}{5} = 75 - 54 + \frac{12}{5} = 21 + \frac{12}{5} = \frac{117}{5}$$

integral

easy

$$\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} dx$$

$$\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} dx = \left[\frac{(2x^3 - 3x^2 - x + 1)^{\frac{4}{3}}}{\frac{4}{3}} \right]_0^1 = -\frac{3}{4} + \frac{3}{4} = 0$$

limit

2026 7th April

derivative

easy

Let $f(x) = \int_0^x \frac{\sin(t)}{1+t^2} dt + \int_x^1 \frac{\cos(t)}{1+t^2} dt$, find $f'(1)$

$$\left(\int_0^x \frac{\sin(t)}{1+t^2} dt \right)' = \frac{\sin(x)}{1+x^2} - 0 = \frac{\sin(x)}{1+x^2},$$

$$\left(\int_x^1 \frac{\cos(t)}{1+t^2} dt \right)' = 1 - \frac{\cos(x)}{1+x^2} = -\frac{\cos(x)}{1+x^2},$$

$$f'(x) = \frac{\sin(x) - \cos(x)}{1+x^2}, \quad f'(1) = \frac{\sin(1) - \cos(1)}{2}$$

medium

Let $f(x) = \frac{e^x \sin(x)}{1+x} + (e - 1 - \frac{1}{10000})x$, find $f'(0)$

$$f'(x) = \frac{(e^x \sin(x) + e^x \cos(x))(1+x) - (e^x \sin(x))}{(1+x)^2} + e - 1 - \frac{1}{10000},$$

$$f'(1) = \frac{1-0}{1} + e - 1 - \frac{1}{10000} = e - \frac{1}{10000}$$

integral

beginner

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2(x) \cos^2(x)} dx$$

$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2(x) \cos^2(x)} dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin^2(x) + \cos^2(x)}{\sin^2(x) \cos^2(x)} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin^2(x)}{\sin^2(x) \cos^2(x)} dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\cos^2(x)}{\sin^2(x) \cos^2(x)} dx = \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\cos^2(x)} dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2(x)} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \csc^2(x) dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2(x) dx = [\tan(x) - \cot(x)]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = \\ &= \left(1 - 1 - \left(\frac{\sqrt{3}}{2} - \sqrt{3} \right) \right) = \frac{2\sqrt{3}}{3} \end{aligned}$$

easy

$$\int_0^1 \left[x^x (\ln x + 1) + \cos(x^2) \cdot 2x + \frac{1}{1+x^2} \right] dx$$

$$\underbrace{\int_0^1 x^x (\ln x + 1) dx}_A + \underbrace{\int_0^1 \cos(x^2) \cdot 2x dx}_B + \underbrace{\int_0^1 \frac{1}{1+x^2} dx}_C,$$

$$y = x^x, \quad \ln(y) = \ln(x^x), \quad \ln(y) = x \ln(x),$$

$$\frac{y'}{y} = x' \ln(x) + x \frac{1}{x}, \quad \frac{y'}{y} = \ln(x) + 1, \quad y' = y \ln(x) + 1, \quad y' = x^x (\ln(x) + 1), \implies A = [x^x]_0^1$$

$$u = x^2, \quad \frac{du}{dx} = 2x, \quad dx = \frac{du}{2x},$$

$$\int_0^1 \cos(u) du = [\sin(u)]_0^1 \implies B = [\sin(x^2)]_0^1$$

$$\int_0^1 \frac{1}{1+x^2} dx = [\arctan(x)]_0^1 = C,$$

$$I = [x^x + \sin(x^2) + \arctan(x)]_0^1 = (1 + \sin(1) + \arctan(1)) - (0 + 1 + 0) = \sin(1) + \arctan(1)$$

2026 8th April

derivative

easy

Let $y(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$, find $y'(-1) - y''(-1)$

$$(1-x)y(x) = (1+x)(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16}) \implies (1-x)y(x) = 1-x^{32} \implies y(x) = \frac{1-x^{32}}{1-x}$$

$$y'(x) = \frac{-32x^{31}(1-x) + (1-x^{32})}{(1-x)^2}, \quad y'(-1) = \frac{64}{4} = 16$$

$$y'(x) = \frac{-32x^{31} + 32x^{32} + 1 - x^{32}}{(1-x)^2},$$

$$y''(x) = \frac{(-992x^{30} + 1024x^{31} - 32x^{31})(1-x)^2 + (-32x^{31} + 32x^{32} + 1 - x^{32})(2-2x)}{(1-x)^4},$$

$$y''(-1) = \frac{(-992 - 1024 + 32)4 + (32 + 32 + 1 - 1)4}{16} = \frac{-1984 + 64}{4} = -\frac{1920}{4} = -480,$$

$$16 - (-480) = 496$$

medium

Let $f(x) = \ln(x + \sqrt{x^2 + 4}) - \frac{x}{\sqrt{x^2 + 4}}$, find $f'(2)$

$$f'(x) = \frac{1 + \frac{x}{\sqrt{x^2+4}}}{x + \sqrt{x^2+4}} - \frac{\sqrt{x^2+4} - \frac{x^2}{\sqrt{x^2+4}}}{x^2+4}$$

$$f'(2) = \frac{1 + \frac{2}{\sqrt{8}}}{2 + \sqrt{8}} - \frac{\sqrt{8} - \frac{4}{\sqrt{8}}}{8}$$

integral

beginner

$$\int_0^1 \frac{\arcsin(x)}{\sqrt{1-x^2}} dx$$

$$\int_0^1 \arcsin(x) \frac{1}{\sqrt{1-x^2}} dx = \int_0^1 \arcsin(x) (\arcsin(x))' dx,$$

$$u = \arcsin(x), \quad \frac{du}{dx} = \frac{1}{\sqrt{1-x^2}}, \quad dx = \sqrt{1-x^2} du,$$

$$\int_0^{\frac{\pi}{2}} u du = \left[\frac{1}{2} u^2 \right]_0^{\frac{\pi}{2}} = \frac{\pi^2}{8}$$

easy

$$\int_0^1 \left(\frac{x \sqrt{\arctan(x)}}{\sqrt{1+x^2}} \right)^2 dx$$

$$\int_0^1 \frac{x^2 \arctan(x)}{1+x^2} dx$$

$$u = \arctan(x), \quad \frac{du}{dx} = \frac{1}{1+x^2}, \quad dx = (1+x^2) du, \quad x = \tan(u)$$

$$\int_0^{\frac{\pi}{4}} \tan^2(u) u \, du = \int_0^{\frac{\pi}{4}} u (\sec^2(u) - 1) \, du = \int_0^{\frac{\pi}{4}} u \sec^2(u) \, du - \int_0^{\frac{\pi}{4}} u \, du,$$

	D	I
+	u	$\sec^2(u)$
−	1	$\tan(u)$
+	0	$-\ln \cos(u) $

$$\left[u \tan(u) + \ln|\cos(u)| - \frac{1}{2}u^2 \right]_0^{\frac{\pi}{4}} = \left(\frac{\pi}{4}(1) + \ln\left(\frac{\sqrt{2}}{2}\right) - \frac{1}{2}\left(\frac{\pi}{4}\right)^2 \right) - (0 + \ln(1) - 0) = \frac{\pi}{4} - \frac{1}{2}\ln(2) - \frac{\pi^2}{32}$$

2026 9th April

derivative

easy

Let $f(x) = \ln(x^2 + e^x) + \arctan(\sqrt{x})$, find $g'(f(1))$, where g is the inverse of f

$$g(f(x)) = x,$$

$$g'(f(x)) \cdot f'(x) = 1,$$

$$g'(f(x)) = \frac{1}{f'(x)} \implies g'(f(1)) = \frac{1}{f'(1)},$$

$$f'(x) = \frac{2x + e^x}{x^2 + e^x} + \frac{1}{(1+x)2\sqrt{x}} = \frac{2x + e^x}{x^2 + e^x} + \frac{1}{2\sqrt{x} + 2x},$$

$$f'(1) = \frac{2+e}{1+e} + \frac{1}{4} \implies \frac{9+5e}{4+4e}$$

$$\frac{1}{f'(1)} = \frac{4+4e}{9+5e}$$

medium

Let $f(x) = \cos(\cos(\cos(\dots(x)\dots)))$, find $f'(\pi)$

$$y = \cos(\cos(\cos(\dots(x)\dots))) \implies y = \cos(y),$$

$$y' = -\sin(y) \cdot y',$$

$$y' + \sin(y) \cdot y' = 0,$$

$$\underbrace{y'(1 + \sin(y))}_{\neq 0} = 0 \implies y' = 0$$

integral

beginner

$$\int_0^1 \frac{x \ln(x^2+1)}{x^2+1} dx$$

$$u = x^2 + 1, \quad \frac{du}{dx} = 2x, \quad dx = \frac{du}{2x}$$

$$\int_1^2 \frac{x \ln(u)}{u} \cdot \frac{1}{2x} du = \int_1^2 \frac{\ln(u)}{2u} du = \frac{1}{2} \int_1^2 \ln(u) \cdot \frac{1}{u} du = \frac{1}{2} \left[\frac{1}{2} \ln^2(u) \right]_1^2 = \frac{1}{4} \ln^2(2)$$

easy

$$\int_1^{\infty} \frac{264}{x(x^4+66)} dx$$

$$264 \int_1^{\infty} \frac{1}{x(x^4+66)} dx = 264 \int_1^{\infty} \frac{x^3}{x^4(x^4+66)} dx,$$

$$u = x^4, \quad \frac{du}{dx} = 4x^3, \quad dx = \frac{1}{4x^3} du,$$

$$264 \lim_{t \rightarrow \infty} \int_1^t \frac{x^3}{u(u+66)} \cdot \frac{1}{4x^3} du = \frac{264}{4} \lim_{t \rightarrow \infty} \int_1^t \frac{1}{u(u+66)} du,$$

$$\frac{A}{u} + \frac{B}{u+66} = 1 \Rightarrow Au + 66A + Bu = 1 \Rightarrow u(A+B) + 66A = 1 \Rightarrow A = \frac{1}{66}, \quad B = -\frac{1}{66},$$

$$\frac{264}{4} \lim_{t \rightarrow \infty} \left(\int_1^t \frac{\frac{1}{66}}{u} du - \int_1^t \frac{\frac{1}{66}}{u+66} du \right) = \lim_{t \rightarrow \infty} [\ln|u| - \ln|u+66|]_1^t = \lim_{t \rightarrow \infty} \left[\ln \left| \frac{u}{u+66} \right| \right]_1^t = -\ln \left(\frac{1}{67} \right)$$

2026 10th April

shit i forgor xdd

2026 11th April

derivative

easy

Let $f(x) = \frac{\sqrt{1+\sin(x^2)}}{1+\cos(x^2)}$, find $f'(1)$

$$f(x) = \frac{\sqrt{1+\sin(x^2)}}{1+\cos(x^2)},$$

$$u(x) = \sqrt{1+\sin(x^2)} \Rightarrow u'(x) = \frac{x \cos(x^2)}{\sqrt{1+\sin(x^2)}},$$

$$v(x) = 1 + \cos(x^2) \Rightarrow v'(x) = -2x \sin(x^2),$$

$$f'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{v(x)^2} = \frac{\frac{x \cos(x^2)}{\sqrt{1+\sin(x^2)}}(1+\cos(x^2)) - (\sqrt{1+\sin(x^2)})(-2x \sin(x^2))}{(1+\cos(x^2))^2},$$

$$f'(1) = \frac{\frac{\cos(1)}{\sqrt{1+\sin(1)}}(1+\cos(1)) - (\sqrt{1+\sin(1)})(-2 \sin(1))}{(1+\cos(1))^2},$$

$$f'(1) = \frac{\cos(1)(1+\cos(1)) + 2 \sin(1)(1+\sin(1))}{\sqrt{1+\sin(1)}(1+\cos(1))^2},$$

$$f'(1) = \frac{\cos(1) + \cos^2(1) + 2 \sin(1) + 2 \sin^2(1)}{\sqrt{1+\sin(1)}(1+\cos(1))^2},$$

$$f'(1) = \frac{\cos(1) + 2 \sin(1) + 1 + \sin^2(1)}{\sqrt{1+\sin(1)}(1+\cos(1))^2}$$

medium

Let $f(x) = \arctan\left(\sqrt{\frac{1-x}{1+x}}\right)$, find $f'\left(\frac{1}{2}\right)$

$$\left(\frac{1-x}{1+x}\right)' = \frac{-1(1+x) - (1-x)}{(1+x)^2} = \frac{-1-x-1+x}{(1+x)^2} = -\frac{2}{(1+x)^2}$$

$$\left(\sqrt{\frac{1-x}{1+x}}\right)' = \frac{1}{2\sqrt{\frac{1-x}{1+x}}} \left(-\frac{2}{(1+x)^2}\right) = -\frac{1}{\sqrt{\frac{1-x}{1+x}}(1+x)^2} = -\frac{\sqrt{1+x}}{\sqrt{1-x}(1+x)^2}$$

$$f'(x) = \frac{1}{1+\left(\sqrt{\frac{1-x}{1+x}}\right)^2} \left(-\frac{\sqrt{1+x}}{\sqrt{1-x}(1+x)^2}\right) = \frac{1+x}{2} \left(-\frac{\sqrt{1+x}}{\sqrt{1-x}(1+x)^2}\right) = -\frac{1}{2\sqrt{1-x}\sqrt{1+x}}$$

$$f'\left(\frac{1}{2}\right) = -\frac{1}{2\sqrt{\frac{1}{2}}\sqrt{\frac{3}{2}}}$$

integral

beginner

$$\int_0^1 \frac{1}{\sqrt{1+x^2}+x} dx$$

$$u = \sqrt{1+x^2}, \quad \frac{du}{dx} = \frac{2x}{2\sqrt{1+x^2}} = \frac{x}{\sqrt{1+x^2}}, \quad dx = \frac{\sqrt{1+x^2}}{x} du, \quad x = \sqrt{u^2-1},$$

$$\int_1^{\sqrt{2}} \left(\frac{1}{u + \sqrt{u^2-1}} \right) \left(\frac{u}{\sqrt{u^2-1}} \right) du = \int_0^{\sqrt{2}} \frac{\sqrt{u^2-1} + u^2 \sqrt{u^2-1}}{u \sqrt{u^2-1} (u^2-1)} du = \int_0^{\sqrt{2}} \frac{1+u^2}{u^3-u} du,$$

this is NOT the way

$$\begin{aligned} & \int_0^1 \left(\frac{1}{\sqrt{1+x^2}+x} \right) \left(\frac{\sqrt{1+x^2}-x}{\sqrt{1+x^2}-x} \right) dx = \int_0^1 \frac{\sqrt{1+x^2}-x}{1+x^2-x^2} dx = \int_0^1 \sqrt{1+x^2} - x dx = \\ & = \int_0^1 \sqrt{1+x^2} dx - \int_0^1 x dx = \left[\frac{1}{2} \left(x\sqrt{1+x^2} + \ln(x + \sqrt{1+x^2}) - x^2 \right) \right]_0^1 = \frac{1}{2} (\sqrt{2} + \ln(1 + \sqrt{2}) - 1) \end{aligned}$$

easy

$$\int_0^1 \frac{e^{3u}}{e^u+1} du$$

$$x = e^u, \quad \frac{dx}{du} = e^u, \quad du = \frac{dx}{e^u}$$

$$\begin{aligned} & \int_1^e \left(\frac{x^3}{x+1} \right) \left(\frac{1}{x} \right) dx = \int_1^e \frac{x^2}{x+1} dx = \int_1^e \frac{x^2+1-1}{x+1} dx = \int_1^e \frac{(x-1)(x+1)+1}{x+1} dx = \\ & = \int_1^e \frac{(x-1)(x+1)}{x+1} dx + \int_1^e \frac{1}{x+1} dx = \left[\frac{x^2}{2} - x + \ln|x+1| \right]_1^e = \frac{e^2}{2} - e + \ln(e+1) - \left(\frac{1}{2} - 1 + \ln(1+1) \right) = \\ & = \frac{e^2}{2} - e + \ln(e+1) + \frac{1}{2} - \ln(2) \end{aligned}$$

2026 12th April

derivative

easy

Let $f(x) = x\sqrt{1-x^2} + \arcsin(x)$, find $f'(\frac{1}{2})$

$$f'(x) = \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}},$$
$$f'\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2} - \frac{\frac{1}{4}}{\frac{\sqrt{3}}{2}} + \frac{1}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{3}}{2} + \frac{\frac{3}{4}}{\frac{\sqrt{3}}{2}}$$

medium

Let $f(x) = e^{x\sqrt{x}}$, find $f''(1)$

$$f'(x) = \frac{3\sqrt{x^3}e^{\sqrt{x^3}}}{2x} = \frac{3\sqrt{x}e^{\sqrt{x^3}}}{2},$$
$$f''(x) = \left(\frac{3}{4\sqrt{x}}\right)(e^{\sqrt{x^3}}) + \left(\frac{3}{2}\sqrt{x}\right)\left(\frac{3}{2}\sqrt{x}e^{\sqrt{x^3}}\right),$$
$$f''(1) = \frac{3}{4}e + \frac{9}{4}e = 3e$$

integral

beginner

$\int_0^{\frac{\pi}{2}} e^{2x}(2\sin(x) + \cos(x)) dx$

$$\int_0^{\frac{\pi}{2}} e^{2x}(2\sin(x) + \cos(x)) dx = \int_0^{\frac{\pi}{2}} e^{2x} \sin(2x) dx,$$

$$u = 2x, \quad \frac{du}{dx} = 2, \quad dx = \frac{1}{2} du,$$

$$\int_0^{\pi} e^u \sin(u) \frac{1}{2} du = \frac{1}{2} \int_0^{\pi} e^u \sin(u) du,$$

	D	I
+	e^u	$\sin(u)$
-	e^u	$-\cos(u)$
+	e^u	$-\sin(u)$

$$I = \frac{1}{2}(e^u(-\cos(u)) + e^u(\sin(u)) - I) = \frac{1}{2}e^u(\sin(u) - \cos(u)) - \frac{1}{2}I,$$

$$\frac{3}{2}I = \frac{1}{2}e^u(\sin(u) - \cos(u)),$$

$$I = \frac{1}{3}e^u(\sin(u) - \cos(u)),$$

$$\left[\frac{1}{3}e^u(\sin(u) - \cos(u))\right]_0^{\pi} = \frac{1}{3}e^{\pi} - \frac{1}{3},$$

guess fucking what turns out fucking $2\sin(x) + \cos(x)$ is not $\sin(2x)$ fuck

$$\int_0^{\frac{\pi}{2}} e^{2x} (2 \sin(x) + \cos(x)) \, dx = [e^{2x} \sin(x)]_0^{\frac{\pi}{2}} = e^{\pi}$$

easy

$$\int_0^1 \frac{2x}{1+\sqrt{3x^2+1}} \, dx$$

$$u = \sqrt{3x^2+1}, \quad \frac{du}{dx} = \frac{3x}{\sqrt{3x^2+1}}, \quad dx = \frac{\sqrt{3x^2+1}}{3x} du,$$

$$\begin{aligned} \int_1^2 \left(\frac{2x}{1+u} \right) \left(\frac{u}{3x} \right) du &= \int_1^2 \frac{2u}{3(1+u)} du = \frac{2}{3} \int_1^2 \frac{u}{1+u} du = \frac{2}{3} \int_1^2 \frac{1+u-1}{1+u} du = \\ &= \frac{2}{3} \left(\int_1^2 1 \, du - \int_1^2 \frac{1}{1+u} \, du \right) = \frac{2}{3} [u - \ln|1+u|]_1^2 = \frac{2}{3} (2 - \ln(3) - (1 - \ln(2))) = \frac{2}{3} \left(1 + \ln\left(\frac{2}{3}\right) \right) \end{aligned}$$

medium

$$\int_{\frac{\sqrt{2}-26}{2}}^{\frac{\sqrt{3}-26}{2}} \arctan\left(\frac{x+13}{\sqrt{-x^2-26x-168}}\right) \, dx$$

$$\int_{\frac{\sqrt{2}-26}{2}}^{\frac{\sqrt{3}-26}{2}} \arctan\left(\frac{x+13}{\sqrt{-(x+12)(x+14)}}\right) \, dx, \text{ hell naw}$$

$$-x^2 - 26x - 168 = -x^2 - 26x - 168 - 1 + 1 = -((x+13)^2 - 1) = 1 - (x+13)^2,$$

$$\int_{\frac{\sqrt{2}-26}{2}}^{\frac{\sqrt{3}-26}{2}} \arctan\left(\frac{x+13}{\sqrt{1-(x+13)^2}}\right) \, dx,$$

$$u = x + 13, \quad \frac{du}{dx} = 1, \quad dx = du,$$

$$\frac{\sqrt{2}-26}{2} + \frac{26}{2} = \frac{\sqrt{2}}{2}, \quad \frac{\sqrt{3}-26}{2} + \frac{26}{2} = \frac{\sqrt{3}}{2}$$

$$\int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \arctan\left(\frac{u}{\sqrt{1-u^2}}\right) \, du,$$

$$\sin(\theta) = u \implies \theta = \arcsin(u), \quad \tan(\theta) = \frac{u}{\sqrt{1-u^2}},$$

$$\begin{aligned} \int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \arctan(\tan(\theta)) \, du &= \int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \arcsin(\theta) \, du = \left[u \arcsin(u) + \sqrt{1-u^2} \right]_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} = \\ &= \left(\frac{\sqrt{3}}{2} \frac{\pi}{3} + \frac{1}{2} \right) - \left(\frac{\sqrt{2}}{2} \frac{\pi}{4} + \frac{\sqrt{2}}{2} \right) = \frac{\sqrt{3}\pi}{6} + \frac{1}{2} - \frac{\sqrt{2}\pi}{8} - \frac{\sqrt{2}}{2} = \frac{4\pi\sqrt{3} + 12 - 3\pi\sqrt{2} - 12\sqrt{2}}{24} \end{aligned}$$

2026 13th April

derivative

easy

Let $f(x) = x^{\frac{\ln(\ln x)}{\ln x}}$, find $f'(e)$

$$f'(x) = \frac{d}{dx} \left(\frac{\ln(\ln x)}{\ln x} \right) = \frac{\frac{1}{x \ln x} \ln x - \ln(\ln x) \frac{1}{x}}{\ln^2 x} = \frac{\frac{1}{x} - \frac{\ln(\ln x)}{x}}{\ln^2 x},$$

$$f'(e) = \frac{\frac{1}{e} - \frac{0}{e}}{\ln^2 e} = \frac{\frac{1}{e}}{1} = \frac{1}{e}$$

medium

Let $f(x) = \frac{\sin(2x) \cosh(x) + \ln(1+x) \cos(x)}{1+x^2}$, find $f'(1)$

$$\begin{aligned} f'(x) &= \\ &= \frac{\left[2 \cos(2x) \cosh(x) + \sin(2x) \sinh(x) + \frac{\cos(x)}{1+x} + \ln(1+x)(-\sin(x)) \right] (1+x^2) - [\sin(2x) \cosh(x) + \ln(1+x) \cos(x)] (2x)}{(1+x^2)^2}, \\ f'(1) &= \frac{2 \cos(2) \cosh(1) + \sin(2) \sinh(1) + \frac{\cos(1)}{2} - \ln(2) \sin(1) - \sin(2) \cosh(1) - \ln(2) \cos(1)}{2} \end{aligned}$$

integral

beginner

$$\int_0^1 \frac{\arctan(\sqrt{x})}{\sqrt{x}(1+x)} dx$$

$$u = \sqrt{x}, \quad \frac{du}{dx} = \frac{1}{2\sqrt{x}}, \quad dx = 2\sqrt{x} du, \quad x = u^2,$$

$$\int_0^1 \frac{\arctan(u)}{u(1+u^2)} \cdot 2u du = 2 \int_0^1 \frac{\arctan(u)}{1+u^2} du = 2 \int_0^1 \arctan(u) \cdot \frac{1}{1+u^2} = 2 \frac{1}{2} [\arctan^2(u)]_0^1 = \arctan^2(1)$$

easy

$$\int_0^1 \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} \, dx$$

$$u = \sqrt{x}, \quad \frac{du}{dx} = \frac{1}{2\sqrt{x}}, \quad dx = 2\sqrt{x} \, du, \quad x = u^2,$$

$$\int_0^1 \sqrt{\frac{1-u}{1+u}} \cdot 2u \, du,$$

$$u = \cos(\theta), \quad \frac{du}{d\theta} = -\sin(\theta), \quad d\theta = -\sin(\theta) \, du,$$

$$2 \int_0^{\frac{\pi}{2}} \sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}} (\cos(\theta)) (-\sin(\theta)) \, d\theta,$$

$$\sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}} = \sqrt{\frac{(1-\cos(\theta))(1-\cos(\theta))}{(1+\cos(\theta))(1-\cos(\theta))}} = \sqrt{\frac{(1-\cos(\theta))^2}{1-\cos^2(\theta)}} = \frac{1-\cos(\theta)}{\sin(\theta)},$$

$$\begin{aligned} 2 \int_0^{\frac{\pi}{2}} \frac{1-\cos(\theta)}{\sin(\theta)} \cos(\theta) \sin(\theta) \, d\theta &= 2 \int_0^{\frac{\pi}{2}} \cos(\theta) - \cos^2(\theta) \, d\theta = 2 \left[\sin(\theta) - \left(\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right) \right]_0^{\frac{\pi}{2}} = \\ &= 2 \left(1 - \frac{\pi}{4} \right) = 2 - \frac{\pi}{2} \end{aligned}$$

2026 14th April

derivative

easy

Let $f(x) = x \sin(x)$, find $f''(\frac{\pi}{4})$

$$f'(x) = \sin(x) + x \cos(x),$$

$$f''(x) = \cos(x) + \cos(x) + x(-\sin(x)) = 2 \cos(x) - x \sin(x)$$

$$f''\left(\frac{\pi}{4}\right) = 2 \frac{\sqrt{2}}{2} - \frac{\pi}{4} \frac{\sqrt{2}}{2} = \sqrt{2} - \frac{\pi\sqrt{2}}{8}$$

medium

Find $\frac{d^{17}}{dx^{17}}(e^x \sin(x))|_{x=0}$

$$f'(x) = e^x \cos(x) + e^x \sin(x),$$

$$f''(x) = -e^x \sin(x) + e^x \cos(x) + e^x \cos(x) + e^x \sin(x) = 2e^x \cos(x),$$

$$f'''(x) = -\sin(x)2e^x + 2e^x \cos(x),$$

$$f^{(4)}(x) = -\cos(x)2e^x - \sin(x)2e^x - 2e^x \sin(x) + 2e^x \cos(x) = 4e^x \sin(x),$$

...

$$f^{(16)}(x) = 256e^x \sin(x),$$

$$f^{(17)}(x) = 256e^x \cos(x) + 256e^x \sin(x) = 256e^x (\cos(x) + \sin(x)),$$

$$f^{(17)}(0) = 256$$

integral

beginner

$$\int_0^{\frac{\pi}{2}} \cos(x) e^{\sqrt{\sin(x)}} dx$$

$$u = \sin(x), \quad \frac{du}{dx} = \cos(x), \quad dx = \frac{1}{\cos(x)} du,$$

$$\int_0^1 \cos(x) e^{\sqrt{u}} \frac{1}{\cos(x)} du = \int_0^1 e^{\sqrt{u}} du,$$

$$v = \sqrt{u}, \quad \frac{dv}{du} = \frac{1}{2\sqrt{u}}, \quad du = 2\sqrt{u} dv,$$

$$\int_0^1 e^v 2v dv = 2 \int_0^1 e^v v dv,$$

	D	I
$+$	v	e^v
$-$	1	e^v
$+$	0	e^v

$$I = e^v v - e^v = e^v (v - 1),$$

$$2[e^v (v - 1)]_0^1 = 2[-(1(-1))] = 2$$

easy

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)}{\tan(x) \sec(x)} \, dx$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)}{\frac{\sin(x)}{\cos(x)} \frac{1}{\cos(x)}} \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)}{\frac{\sin(x)}{\cos^2(x)}} \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos(x) \frac{\cos^2(x)}{\sin(x)} \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos(x) \frac{1 - \sin^2(x)}{\sin(x)} \, dx,$$

$$u = \sin(x), \quad \frac{du}{dx} = \cos(x), \quad dx = \frac{1}{\cos(x)} \, du,$$

$$\int_{\frac{\sqrt{2}}{2}}^1 \cos(x) \frac{1}{\cos(x)} \frac{1 - u^2}{u} \, du = \int_{\frac{\sqrt{2}}{2}}^1 \frac{1 - u^2}{u} \, du = \int_{\frac{\sqrt{2}}{2}}^1 \frac{1}{u} \, du - \int_{\frac{\sqrt{2}}{2}}^1 u \, du = \left[\ln|u| - \frac{u^2}{2} \right]_{\frac{\sqrt{2}}{2}}^1 =$$

$$= -\frac{1}{2} - \left(\ln\left(\frac{\sqrt{2}}{2}\right) - \frac{\frac{1}{2}}{2} \right) = -\frac{1}{2} - \ln\left(\frac{\sqrt{2}}{2}\right) + \frac{1}{4} = -\frac{1}{4} - \ln\left(\frac{\sqrt{2}}{2}\right)$$

2026 15th April

derivative

easy

Find the slope $(y - x^5)^2 = x(1 + x^2)^2$ at the $(1, 3)$ point.

$$\partial_x = 2(y - x^5)(-5x^4) = (1 + x^2)^2 + 2(1 + x^2)(2x) \implies$$

$$\implies 2(y - x^5)(-5x^4) - (1 + x^2)^2 - 2(1 + x^2)(2x) = 0$$

$$\partial_y = 2(y - x^5) = 0$$

$$\partial_x f(a) = 2(3 - 1)(-5) - (1 + 1)^2 - 2(1 + 1)(2) = -20 - 4 - 8 = -32$$

$$\partial_y f(a) = 2(3 - 1) = 4$$

$$\frac{dy}{dx} = -\frac{\partial_x f(a)}{\partial_y f(a)} = -\frac{-32}{4} = 8$$

medium

Let $f(x) = \sin\left(\frac{e^{\tan(x)}}{\ln(x)}\right)$, find $f'(e)$

$$f'(x) = \cos\left(\frac{e^{\tan(x)}}{\ln(x)}\right) \left(\frac{\sec^2(x) e^{\tan(x)} \ln(x) - \frac{e^{\tan(x)}}{x}}{\ln^2(x)} \right) = \cos\left(\frac{e^{\tan(x)}}{\ln(x)}\right) \left(\frac{e^{\tan(x)} (\sec^2(x) \ln(x) - \frac{1}{x})}{\ln^2(x)} \right),$$

$$f'(e) = \cos(e^{\tan(e)}) e^{\tan(e)} \left(\sec^2(e) - \frac{1}{e} \right)$$

integral

beginner

$$\int_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} e^{\sin(x)} e^{\cos(x)} (\cos(x) - \sin(x)) dx$$

$$\int_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} e^{\sin(x) + \cos(x)} (\cos(x) - \sin(x)) dx,$$

$$u = \sin(x) + \cos(x), \quad \frac{du}{dx} = \cos(x) - \sin(x), \quad dx = \frac{1}{\cos(x) - \sin(x)},$$

$$\sin\left(\frac{5\pi}{4}\right) + \cos\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} + -\frac{\sqrt{2}}{2} = -\sqrt{2}$$

$$\sin\left(\frac{9\pi}{4}\right) + \cos\left(\frac{9\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$$

$$\int_{-\sqrt{2}}^{\sqrt{2}} e^u (\cos(x) - \sin(x)) \frac{1}{\cos(x) - \sin(x)} du = \int_{-\sqrt{2}}^{\sqrt{2}} e^u du = [e^u]_{-\sqrt{2}}^{\sqrt{2}} = e^{\sqrt{2}} - e^{-\sqrt{2}}$$

easy

$\int_0^\pi x \cos^2(x) \, dx$

	D	I
$+$	x	$\cos^2(x)$
$-$	1	$\frac{x}{2} + \frac{\sin(2x)}{4}$
$+$	0	$\frac{x^2}{4} - \frac{\cos(2x)}{8}$

$$\int_0^\pi x \cos^2(x) \, dx = \left[x \left(\frac{x}{2} + \frac{\sin(2x)}{4} \right) - \left(\frac{x^2}{4} - \frac{\cos(2x)}{8} \right) \right]_0^\pi = \frac{\pi^2}{2} - \frac{\pi^2}{4} + \frac{1}{8} - \frac{1}{8} = \frac{\pi^2}{4}$$

2026 16th April

derivative

easy

Let $f(1) = 1$ and $f'(1) = 3$, find the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$

$$f(x) = x^3, \quad f'(x) = 3x^2,$$

$$f(f(f(x))) + (f(x))^2 = \left((x^3)^3\right)^3 + (x^3)^2 = x^{27} + x^6,$$

$$\frac{d}{dx}(f(f(f(x))) + (f(x))^2) = 27x^{26} + 6x^5 \stackrel{1}{\Rightarrow} = 27 + 6 = 33$$

medium

Let $f(x) = \sum_{n=1}^{50} nx^n$, find $f'(1)$

$$f(x) = x + 2x^2 + 3x^3 + \dots + 50x^{50}$$

$$f'(x) = \sum_{n=1}^{50} n^2 x^{n-1} = 1 + 4x + 9x^2 + \dots + 2500x^{49}$$

$$f'(1) = \sum_{n=1}^{50} n^2 = \frac{50 \cdot 51 \cdot 101}{6} = 42925$$

integral

beginner

$$\int_0^{\frac{\pi}{2}} \frac{1}{\sec(x) + \tan(x)} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{\frac{1}{\cos(x)} + \frac{\sin(x)}{\cos(x)}} dx = \int_0^{\frac{\pi}{2}} \frac{1}{\frac{1+\sin(x)}{\cos(x)}} dx = \int_0^{\frac{\pi}{2}} \frac{\cos(x)}{1+\sin(x)} dx,$$

$$u = \sin(x), \quad \frac{du}{dx} = \cos(x), \quad dx = \frac{1}{\cos(x)} du,$$

$$\int_0^1 \frac{1}{1+u} du = [\ln|1+u|]_0^1 = \ln(2)$$

easy

$$\int_0^1 \frac{\arctan(x) - x \arctan(x)}{1-x+x^2-x^3} dx$$

$$1-x+x^2-x^3 = (1-x) + x^2(1-x) = (1-x)(1+x^2)$$

$$\int_0^1 \frac{\arctan(x) - x \arctan(x)}{1-x+x^2-x^3} dx = \int_0^1 \frac{\arctan(x)(1-x)}{(1-x)(1+x^2)} dx = \int_0^1 \frac{\arctan(x)}{1+x^2} dx =$$

$$= \frac{1}{2} [\arctan^2(x)]_0^1 = \frac{\arctan^2(1)}{2}$$

2026 17th April

derivative

easy

Find the maximum value of $\frac{1}{a}$ such that the inequality $4ax^2 + \frac{1}{x} \geq 1$ holds for all $x > 0$

$$4ax^2 \geq 1 - \frac{1}{x}$$

$$a \geq \frac{1 - \frac{1}{x}}{4x^2}$$

$$a \geq \frac{x - 1}{4x^3}$$

$$\frac{d}{dx} \left(\frac{x - 1}{4x^3} \right) = \frac{4x^3 - (x - 1)(12x^2)}{(4x^3)^2} = \frac{4x^3 - 12x^3 + 12x^2}{16x^6} = \frac{12x^2 - 8x^3}{16x^6} = \frac{3 - 2x}{4x^4},$$

$$\frac{3 - 2x}{4x^4} = 0 \iff 3 = 2x \iff x = \frac{3}{2},$$

$$\frac{\frac{3}{2} - 1}{4\left(\frac{3}{2}\right)^3} = \frac{\frac{1}{2}}{4\frac{27}{8}} = \frac{1}{2} \cdot \frac{2}{27} = \frac{1}{27}$$

$$a = 27$$

medium

Let $f(x) = \arctan\left(\frac{x}{1+\sqrt{1+x^2}}\right)$, find $f'(\sqrt{2})$

$$f'(x) = \frac{1}{1 + \left(\frac{x}{1+\sqrt{1+x^2}}\right)^2} \cdot \frac{1 + \sqrt{1+x^2} - \frac{x^2}{\sqrt{1+x^2}}}{(1 + \sqrt{1+x^2})^2},$$

$$\begin{aligned} f'(\sqrt{2}) &= \frac{1}{1 + \left(\frac{\sqrt{2}}{1+\sqrt{3}}\right)^2} \cdot \frac{1 + \sqrt{3} - \frac{2}{\sqrt{3}}}{(1 + \sqrt{3})^2} = \frac{1}{1 + \frac{2}{1+2\sqrt{3}+3}} \cdot \frac{1 + \sqrt{3} - \frac{2}{\sqrt{3}}}{1 + 2\sqrt{3} + 3} = \\ &= \frac{1 + \sqrt{3} - \frac{2}{\sqrt{3}}}{1 + 2\sqrt{3} + 3 + 2} = \frac{1 + \sqrt{3} - \frac{2}{\sqrt{3}}}{6 + 2\sqrt{3}} \end{aligned}$$

integral

beginner

$$\int_2^e \frac{\ln(\ln(x))}{x \ln(x)} dx$$

$$u = \ln(x), \quad \frac{du}{dx} = \frac{1}{x}, \quad dx = x du,$$

$$\int_{\ln(2)}^1 \frac{\ln(u)}{u} du = \frac{1}{2} [\ln^2(u)]_{\ln(2)}^1 = -\frac{1}{2} \ln^2(\ln(2))$$

easy

$$\int_1^2 \frac{2x^2(x+1)-3}{x(x^3+2x^2+5x+3)} dx$$

$$\frac{d}{dx}(x^4 + 2x^3 + 5x^2 + 3x) = 4x^3 + 6x^2 + 10x + 3$$

$$\int_1^2 \frac{2x^3 + 2x^2 - 3}{x^4 + 2x^3 + 5x^2 + 3x} dx = \int_1^2 \frac{(4x^3 + 6x^2 + 10x + 3) - 2x^3 - 4x^2 - 10x - 6}{x^4 + 2x^3 + 5x^2 + 3x} dx,$$

what a bullshit question wow

$$\int_1^2 \frac{2x^3 + 2x^2 - 3}{x^4 + 2x^3 + 5x^2 + 3x} dx = \int_1^2 \frac{2x + 2 - \frac{3}{x^2}}{x^2 + 2x + 5 + \frac{3}{x}}$$

$$\frac{d}{dx}\left(x^2 + 2x + 5 + \frac{3}{x}\right) = 2x + 2 + \frac{x-3}{x^2} = 2x + 2 - \frac{3}{x^2}$$

$$\int_1^2 \frac{\frac{d}{dx}\left(x^2 + 2x + 5 + \frac{3}{x}\right)}{x^2 + 2x + 5 + \frac{3}{x}} = \left[\ln \left| x^2 + 2x + 5 + \frac{3}{x} \right| \right]_1^2 = \ln\left(\frac{29}{2}\right) - \ln(11) = \ln\left(\frac{29}{22}\right)$$

2026 18th April

derivative

easy

Let f be a real valued differentiable function on $\mathbb{R}$ such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to.

$$Y - y = \frac{dy}{dx}(0 - x)$$

$$Y - y = -x \frac{dy}{dx}$$

$$Y = y - x \frac{dy}{dx} = x^3$$

$$\frac{dy}{dx} - \frac{1}{x}y = -x^2$$

$$\frac{1}{x} \frac{dy}{dx} - \frac{1}{x^2}y = -x$$

$$\frac{d}{dx} \left(y \cdot \frac{1}{x} \right) = -x$$

$$\frac{y}{x} = -\frac{x^2}{2} + C$$

$$y = -\frac{x^3}{2} + Cx$$

$$1 = -\frac{1^3}{2} + C(1)$$

$$1 = -\frac{1}{2} + C$$

$$C = \frac{3}{2}$$

$$f(x) = -\frac{x^3}{2} + \frac{3x}{2}$$

$$f(-3) = \frac{(-3)^3}{2} + \frac{3(-3)}{2}$$

$$f(-3) = \frac{18}{2} = 9$$

medium

A curve C has equation

$$2^x + y^2 = 2xy$$

find the exact value of $\frac{dy}{dx}$ at the point on C with coordinates $(3, 2)$.

$$\frac{d}{dx}(2^x + y^2) = 2^x \ln(2) + 2y \frac{dy}{dx} \quad \frac{d}{dx}(2xy) = 2y + 2x \frac{dy}{dx}$$

$$2^x \ln(2) + 2y \frac{dy}{dx} = 2y + 2x \frac{dy}{dx},$$

$$2^3 \ln(2) + 2 \cdot 2 \frac{dy}{dx} = 2 \cdot 2 + 2 \cdot 3 \frac{dy}{dx}$$

$$8 \ln(2) + 4 \frac{dy}{dx} = 4 + 6 \frac{dy}{dx}$$

$$8 \ln(2) = 4 + 2 \frac{dy}{dx}$$

$$4 \ln(2) = 2 + \frac{dy}{dx}$$

$$\frac{dy}{dx} = 4 \ln(2) - 2$$

integral

beginner

$$\int_1^9 \frac{x^2 + \sqrt{x} + 1}{x^2} dx$$

$$\int_1^9 \frac{x^{\frac{1}{2}}}{x^2} dx = \int_1^9 x^{\frac{1}{2}} \cdot x^{-2} dx = \int_1^9 x^{-\frac{3}{2}} dx$$

$$\begin{aligned} \int_1^9 1 dx + \int_1^9 x^{-\frac{3}{2}} dx + \int_1^9 x^{-2} dx &= \left[x + \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + \frac{x^{-1}}{-1} \right]_1^9 = \left[x - \frac{2}{\sqrt{x}} - \frac{1}{x} \right]_1^9 = \left(9 - \frac{2}{3} - \frac{1}{9} \right) - (1 - 2 - 1) = \\ &= \frac{81}{9} - \frac{6}{9} - \frac{1}{9} - 1 + 2 + 1 = \frac{74}{9} + 2 = \frac{92}{9} \end{aligned}$$

easy

$$\int_1^e \ln^3(x) \, dx$$

$$u = \ln(x), \quad \frac{du}{dx} = \frac{1}{x}, \quad dx = x \, du, \quad x = e^u,$$

$$\int_0^1 u^3 e^u \, du$$

	D	I
$+$	u^3	e^u
$-$	$3u^2$	e^u
$+$	$6u$	e^u
$-$	6	e^u
$+$	0	e^u

$$\int_0^1 u^3 e^u \, du = [e^u(u^3 - 3u^2 + 6u - 6)]_0^1 = (e(1 - 3 + 6 - 6)) - (-6) = -2e + 6$$

2026 19th April

derivative

easy

Let $f(x) = x^2 - 3x + 2$ with domain $[1.5, \infty)$, find $[f^{-1}]'(3)$

$$y = x^2 - 3x + 2,$$

$$y - 2 = x(x - 3) \implies \text{not doing ts}$$

$$[f^{-1}(y)] = \frac{1}{f'(x)},$$

$$x^2 - 3x + 2 = 3 \implies x^2 - 3x - 1 = 0,$$

$$x = \frac{3 \pm \sqrt{13}}{2} = \frac{3 + \sqrt{13}}{2},$$

$$f'(x) = 2x - 3,$$

$$f'\left(\frac{3 + \sqrt{13}}{2}\right) = \sqrt{13},$$

$$[f^{-1}]'(3) = \frac{1}{\sqrt{13}}$$

medium

Let $f(x) = \sin(x) + \cos(x) + \tan(x) + \cot(x) + \sec(x) + \csc(x)$, find the minimum value of $(f(x))^2$

$$(f(x))^2 = (\sin(x) + \cos(x) + \tan(x) + \cot(x) + \sec(x) + \csc(x))^2,$$

$$[(f(x))^2]' = 2(\sin(x) + \cos(x) + \tan(x) + \cot(x) + \sec(x) + \csc(x))$$

$$(\cos(x) - \sin(x) + \sec^2(x) - \csc^2(x) + (\sec(x)\tan(x) - (\csc(x)\cot(x))))$$

akkor a kurva anyadat azt

$$\begin{aligned} f(x) &= \sin(x) + \cos(x) + \frac{\sin(x)}{\cos(x)} + \frac{\cos(x)}{\sin(x)} + \frac{1}{\cos(x)} + \frac{1}{\sin(x)} = \\ &= (\sin(x) + \cos(x)) + \underbrace{\frac{\sin^2(x) + \cos^2(x)}{\cos(x)\sin(x)}}_{\frac{1}{\cos(x)\sin(x)}} + \frac{\sin(x) + \cos(x)}{\cos(x)\sin(x)}, \end{aligned}$$

$$u = \sin(x) + \cos(x), \quad u^2 = \sin^2(x) + \cos^2(x) + 2\sin(x)\cos(x) = 1 + 2\sin(x)\cos(x),$$

$$\frac{u^2 - 1}{2} = \sin(x)\cos(x),$$

$$g(u) = u + \frac{1+u}{\frac{u^2-1}{2}} = u + (1+u)\left(\frac{2}{u^2-1}\right) = u + \frac{2+2u}{u^2-1} = u + \frac{2(u+1)}{(u-1)(u+1)} = u + \frac{2}{u-1},$$

$$g'(u) = 1 + \frac{2}{(u-1)^2},$$

$$1 = \frac{2}{(u-1)^2} \implies (u-1)^2 = 2 \implies u-1 = \sqrt{2} \implies u = 1 \pm \sqrt{2},$$

$$g(1 - \sqrt{2}) = 1 - \sqrt{2} + \frac{2}{-\sqrt{2}} = 1 - \sqrt{2} - \sqrt{2} = 1 - 2\sqrt{2},$$

$$(1 - 2\sqrt{2})^2 = 1 - 4\sqrt{2} + 8$$

integral

beginner

$$\int_0^1 \frac{\ln(1+x)}{1+x} dx$$

$$u = x + 1, \quad \frac{du}{dx} = 1, \quad dx = du,$$

$$\int_1^2 \frac{\ln(u)}{u} du = \frac{1}{2} [\ln^2(u)]_1^2 = \frac{1}{2} \ln^2(2)$$

easy

$$\int_0^{\sqrt{\frac{\pi}{4}}} 4x(\sec^2(x^2) + \sec^2(x^2) \tan^2(\tan(x^2))) dx$$

$$u = x^2, \quad \frac{du}{dx} = 2x, \quad dx = \frac{1}{2x} du,$$

$$\int_0^{\frac{\pi}{4}} 2(\sec^2(u) + \sec^2(u) \tan^2(\tan(u))) du = \int_0^{\frac{\pi}{4}} 2(\sec^2(u)(1 + \tan^2(\tan(u)))) du =$$

$$= \int_0^{\frac{\pi}{4}} 2(\sec^2(u)(\sec^2(\tan(u)))) du,$$

$$w = \tan(u), \quad \frac{dw}{du} = \sec^2(u), \quad du = \cos^2(u) dw,$$

$$\int_0^1 2(\sec^2(u) \cdot (\sec^2(w))) \cos^2(u) dw = 2 \int_0^1 \sec^2(w) dw = 2[\tan(w)]_0^1 = 2 \tan(1)$$

2026 20th April

derivative

easy

Let $f(x) = \frac{1+\cos(x^2)}{1+\sin(x^2)}$, find $f'(1)$

$$\begin{aligned} f'(x) &= \frac{(-\sin(x^2)2x)(1+\sin(x^2)) - (1+\cos(x^2))(\cos(x^2)2x)}{(1+\sin(x^2))^2} = \\ &= \frac{-2x\sin(x^2) - 2x\sin^2(x^2) - 2x\cos(x^2) - 2x\cos^2(x^2)}{(1+\sin(x^2))^2} = \frac{-2x(\sin(x^2) + \sin^2(x^2) + \cos(x^2) + \cos^2(x^2))}{(1+\sin(x^2))^2} = \\ &= \frac{-2x(\sin(x^2) + \cos(x^2) + 1)}{1 + 2\sin(x^2) + \sin^2(x^2)}, \\ f'(1) &= \frac{-2\sin(1) - 2\cos(1) - 2}{1 + 2\sin(1) + \sin^2(1)} \end{aligned}$$

medium

Let $f(x) = x^5 + 2e^{\frac{x}{4}}$ for all $x \in \mathbb{R}$. Consider a function $g(x)$ such that $(g \circ f)(x) = x$ for all $x \in \mathbb{R}$. Find the value of $8g'(2)$.

we need: $f(x) = 2$

if $x = 0$:

$$f(x) = 0 + 2e^{\frac{0}{4}} = 0 + 2 \cdot 1 = 2,$$

$$g'(y) = \frac{1}{f'(x)},$$

$$f'(x) = 5x^4 + \frac{4}{16}2e^{\frac{x}{4}} = 5x^4 + \frac{e^{\frac{x}{4}}}{2},$$

$$f'(0) = \frac{1}{2},$$

$$g'(2) = \frac{1}{\frac{1}{2}} = 2,$$

$$8g'(2) = 8 \cdot 2 = 16$$

integral

beginner

$$\int_0^{\frac{\pi}{2}} \sin^3(x) \cos^2(x) \, dx$$

$$\int_0^{\frac{\pi}{2}} \sin(x) \cdot \sin^2(x) \cos^2(x) \, dx,$$

$$u = \cos(x), \quad \frac{du}{dx} = -\sin(x), \quad dx = -\frac{1}{\sin(x)} du,$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin(x) \cdot (1 - \cos^2(x)) \cos^2(x) \, dx &= \int_1^0 -1 \cdot (1 - u^2) u^2 \, du = - \int_1^0 u^2 - u^4 \, du = - \left[\frac{u^3}{3} - \frac{u^5}{5} \right]_1^0 = \\ &= \left[\frac{u^3}{3} - \frac{u^5}{5} \right]_0^1 = \left(\frac{1}{3} - \frac{1}{5} \right) - 0 = \frac{2}{15} \end{aligned}$$

easy

$$\int_0^\infty \frac{\arcsin(\sqrt{1-e^{-2x}})}{\sqrt{e^{2x}-1}} \, dx$$

$$u = \sqrt{1 - e^{-2x}}, \quad \frac{du}{dx} = \frac{e^{-2x}}{\sqrt{1 - e^{-2x}}}, \quad dx = \frac{\sqrt{1 - e^{-2x}}}{e^{-2x}} du, \quad 1 - u^2 = e^{-2x},$$

$$dx = \frac{u}{1 - u^2} du,$$

$$\sqrt{e^{2x} - 1} = \sqrt{\frac{1}{1 - u^2} - 1} = \sqrt{\frac{u^2}{1 - u^2}}$$

$$\begin{aligned} \int_0^1 \frac{\arcsin(u)}{\sqrt{\frac{u^2}{1-u^2}}} \cdot \frac{u}{1-u^2} \, du &= \int_0^1 \frac{\arcsin(u)}{\frac{u}{\sqrt{1-u^2}}} \cdot \frac{u}{1-u^2} \, du = \int_0^1 \frac{\sqrt{1-u^2} \arcsin(u)}{u} \cdot \frac{u}{1-u^2} \, du = \\ &= \int_0^1 \frac{\arcsin(u)}{\sqrt{1-u^2}} \, du = \frac{1}{2} [\arcsin^2(u)]_0^1 = \frac{1}{2} \left[\left(\frac{\pi}{2} \right)^2 - 0^2 \right] = \frac{\pi^2}{8} \end{aligned}$$

2026 21th April

derivative

easy

Let $x^2y + y^3 = 170$, find $\frac{dy}{dx}$ at $(3, 5)$

$$\begin{aligned}\partial_x &= 2xy, & \partial_y &= x^2 + 3y^2, \\ \frac{dy}{dx} &= -\frac{\partial_x}{\partial_y} = \frac{2xy}{x^2 + 3y^2} \Rightarrow \frac{2 \cdot 3 \cdot 5}{9 + 3 \cdot 25} = -\frac{30}{84} = -\frac{5}{14}\end{aligned}$$

medium

Let $y(x) = \begin{vmatrix} \sin(x) & \cos(x) & \sin(x) + \cos(x) + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, find $\frac{d^2y}{dx^2} + y$

$$\begin{aligned}y(x) &= \sin(x) \begin{vmatrix} 28 & 27 \\ 1 & 1 \end{vmatrix} - \cos(x) \begin{vmatrix} 27 & 27 \\ 1 & 1 \end{vmatrix} + (\sin(x) + \cos(x) + 1) \begin{vmatrix} 27 & 28 \\ 1 & 1 \end{vmatrix} = \\ &= \sin(x) + (\sin(x) + \cos(x) + 1)(-1) = -\cos(x) - 1, \\ y' &= \sin(x), & y'' &= \cos(x), \\ \frac{d^2y}{dx^2} + y &= \cos(x) - \cos(x) - 1 = -1\end{aligned}$$

integral

beginner

$\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$

$$\begin{aligned}x &= \sin^2(\theta), & \frac{dx}{d\theta} &= 2 \sin(\theta) \cos(\theta), & dx &= 2 \sin(\theta) \cos(\theta) d\theta, \\ \int_0^{\frac{\pi}{2}} &= \frac{2 \sin(\theta) \cos(\theta)}{\sqrt{\sin^2(\theta)(1 - \sin^2(\theta))}} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{\sin(\theta) \cos(\theta)} 2 \sin(\theta) \cos(\theta) d\theta = \int_0^{\frac{\pi}{2}} 2 d\theta = \\ &= 2[\theta]_0^{\frac{\pi}{2}} = 2\frac{\pi}{2} = \pi\end{aligned}$$

easy

$$\int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left[\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right]^{\frac{1}{2}} dx$$

$$A = \frac{x+1}{x-1}, \quad B = \frac{x-1}{x+1} \implies AB = 1,$$

$$A^2 - 2 \underbrace{AB}_1 + B^2 = (A - B)^2,$$

$$\begin{aligned} & \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left[\left(\frac{x+1}{x-1} - \frac{x-1}{x+1} \right)^2 \right]^{\frac{1}{2}} dx = \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left| \frac{x+1}{x-1} - \frac{x-1}{x+1} \right| dx = \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left| \frac{(x+1)^2 - (x-1)^2}{(x-1)(x+1)} \right| dx = \\ &= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left| \frac{x^2 + 2x + 1 - (x^2 - 2x + 1)}{x^2 - 1} \right| dx = \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left| \frac{4x}{x^2 - 1} \right| dx = \int_{-\frac{1}{\sqrt{2}}}^0 \left| \frac{4x}{x^2 - 1} \right| dx + \int_0^{\frac{1}{\sqrt{2}}} \left| \frac{4x}{x^2 - 1} \right| dx = \\ &= 2 \int_{-\frac{1}{\sqrt{2}}}^0 \frac{2x}{x^2 - 1} dx - 2 \int_0^{\frac{1}{\sqrt{2}}} \frac{2x}{x^2 - 1} dx = 2 [\ln|x^2 - 1|]_{-\frac{1}{\sqrt{2}}}^0 - 2 [\ln|x^2 - 1|]_0^{\frac{1}{\sqrt{2}}} = \\ &= 2 \left(0 - \ln\left(\frac{1}{2}\right) \right) - 2 \left(\ln\left(\frac{1}{2}\right) - 0 \right) = -4 \ln\left(\frac{1}{2}\right) = 4 \ln(2) \end{aligned}$$

2026 22th April

derivative

easy

Let $x = 16 \sin(2t)$ and $y = e^{2t} - \sin(2t)$. Find $\frac{dy}{dx}$ when $t = -67$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{dx}{dt} = 16 \cos(2t) \cdot 2 = 32 \cos(2t)$$

$$\frac{dy}{dt} = e^{2t} \cdot 2 - \cos(2t) \cdot 2 = 2e^{2t} - 2 \cos(2t)$$

$$\frac{dy}{dx} = \frac{2e^{2t} - 2 \cos(2t)}{32 \cos(2t)} = \frac{e^{2t} - \cos(2t)}{16 \cos(2t)}$$

$$\left. \frac{dy}{dx} \right|_{t=-67} = \frac{e^{2(-67)} - \cos(2(-67))}{16 \cos(2(-67))} = \frac{e^{-134} - \cos(-134)}{16 \cos(-134)} = \frac{e^{-134} - \cos(134)}{16 \cos(134)}$$

medium

Let $f(x) = \frac{\sin(\pi x)}{x}$, find $f''(1)$

$$f'(x) = \frac{\cos(\pi x)\pi x - \sin(\pi x)}{x^2}$$

$$f''(x) = \frac{(-\pi^2 x \sin(\pi x))x^2 - (\cos(\pi x)\pi x - \sin(\pi x))2x}{x^4}$$

$$f''(1) = -\pi \sin(\pi) - (\cos(\pi)\pi - \sin(\pi))2 = -2(-\pi) = 2\pi$$

integral

beginner

$$\int_0^{\frac{2\pi}{5}} \sec^3(x) \sin(x) dx$$

$$\int_0^{\frac{2\pi}{5}} \frac{1}{\cos^3(x)} \sin(x) dx = \int_0^{\frac{2\pi}{5}} \tan(x) \sec^2(x) dx = \frac{1}{2} [\tan^2(x)]_0^{\frac{2\pi}{5}} = \frac{1}{2} \tan^2\left(\frac{2\pi}{5}\right)$$

easy

$$\int_1^e x^{\frac{\ln(\ln(x))}{\ln(x)}} dx$$

$$\begin{aligned} \int_1^e (e^{\ln(x)})^{\frac{\ln(\ln(x))}{\ln(x)}} dx &= \int_1^e e^{\ln(x) \cdot \frac{\ln(\ln(x))}{\ln(x)}} dx = \int_1^e e^{\ln(\ln(x))} dx = \int_1^e \ln(x) dx = \\ &= [x \ln(x) - x]_1^e = e - e + 1 = 1 \end{aligned}$$

2026 23th April

derivative

easy

Let $f(x) = \ln(x) + \frac{1}{x}$, find the minimum value of $f(x)$

$$f'(x) = \frac{1}{x} - \frac{1}{x^2} = 0 \implies x = 1$$

medium

Let $f(x) = \ln(\cos^3(\sqrt{x}))$, find $f'\left(\frac{\pi^2}{8}\right)$

$$f'(x) = \frac{3 \cos^2(\sqrt{x})(-\sin(\sqrt{x})) \frac{1}{2\sqrt{x}}}{\cos^3(\sqrt{x})} = -\frac{3 \sin(\sqrt{x})}{\cos(\sqrt{x}) 2\sqrt{x}} = -\frac{3}{2} \cdot \frac{\tan(\sqrt{x})}{\sqrt{x}}$$

$$f'\left(\frac{\pi^2}{8}\right) = -\frac{3}{2} \cdot \frac{\tan\left(\frac{\pi}{2\sqrt{2}}\right)}{\frac{\pi}{2\sqrt{2}}} = -\frac{3}{2} \tan\left(\frac{\pi}{2\sqrt{2}}\right) \frac{2\sqrt{2}}{\pi}$$

integral

beginner

$$\int_0^{\frac{\pi}{4}} \frac{e^{\sqrt{x}} - \sin(\sqrt{x})}{2\sqrt{x}} dx$$

$$u = \sqrt{x}, \quad \frac{du}{dx} = \frac{1}{2\sqrt{x}}, \quad dx = 2\sqrt{x} du,$$

$$\int_0^{\frac{\sqrt{\pi}}{2}} e^u - \sin(u) du = [e^u + \cos(u)]_0^{\frac{\sqrt{\pi}}{2}} = e^{\frac{\sqrt{\pi}}{2}} + \cos\left(\frac{\sqrt{\pi}}{2}\right) - 1 - 1 = e^{\frac{\sqrt{\pi}}{2}} + \cos\left(\frac{\sqrt{\pi}}{2}\right) - 2$$

easy

$$\int_0^{\pi} \frac{1}{1+3^{\cos(x)}} dx$$

$$\begin{aligned} I &= \int_0^{\pi} \frac{1}{1+3^{\cos(x)}} dx = \int_0^{\pi} \frac{1}{1+3^{\cos(\pi-x)}} dx = \int_0^{\pi} \frac{1}{1+3^{-\cos(x)}} dx = \int_0^{\pi} \frac{3^{\cos(x)}}{(1+3^{-\cos(x)})3^{\cos(x)}} dx = \\ &= \int_0^{\pi} \frac{3^{\cos(x)}}{3^{\cos(x)}+1} dx = I \end{aligned}$$

$$2I = \int_0^{\pi} \left(\frac{1}{1+3^{\cos(x)}} + \frac{3^{\cos(x)}}{3^{\cos(x)}+1} \right) dx = \int_0^{\pi} \frac{1+3^{\cos(x)}}{1+3^{\cos(x)}} dx = \int_0^{\pi} 1 dx = [x]_0^{\pi} = \pi$$

$$2I = \pi \implies I = \frac{\pi}{2}$$

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derivative

easy

Let $y = x^{\sin(x)}$, find $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$

$$\ln(y) = \ln(x^{\sin(x)})$$

$$\ln(y) = \ln(x) \sin(x)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\sin(x)}{x} + \ln(x) \cos(x)$$

$$\frac{dy}{dx} = y \left(\frac{\sin(x)}{x} + \ln(x) \cos(x) \right)$$

$$\frac{dy}{dx} = x^{\sin(x)} \left(\frac{\sin(x)}{x} + \ln(x) \cos(x) \right)$$

$$x = \frac{\pi}{2} \Rightarrow \frac{dy}{dx} = \frac{\pi}{2} \left(\frac{2}{\pi} + \ln\left(\frac{\pi}{2}\right) \cdot 0 \right)$$

$$\frac{dy}{dx} = 1$$

medium

Let $f(x) = \frac{|x|}{2 \cos(\ln(x))}$, find $f'(200)$

$$\begin{aligned} f'(x) &\stackrel{200>0}{=} \frac{d}{dx} \left(\frac{x}{2 \cos(\ln(x))} \right) = \frac{2 \cos(\ln(x)) - x(2(-\sin(\ln(x))\frac{1}{x}))}{(2 \cos(\ln(x)))^2} = \frac{2 \cos(\ln(x)) + 2 \sin(\ln(x))}{(2 \cos(\ln(x)))^2} = \\ &= \frac{\cos(\ln(x)) + \sin(\ln(x))}{2 \cos^2(\ln(x))}, \\ f'(200) &= \frac{\cos(\ln(200)) + \sin(\ln(200))}{2 \cos^2(\ln(200))} \end{aligned}$$

integral

beginner

$$\int_0^{\frac{\pi}{2}} \frac{\sin(x)}{1 + \cos^2(x)} dx$$

$$\frac{d}{dx}(1 + \cos^2(x)) = -2 \cos(x) \sin(x),$$

$$u = \cos(x), \quad \frac{du}{dx} = -\sin(x), \quad dx = -\frac{1}{\sin(x)},$$

$$- \int_1^0 \frac{1}{1 + u^2} du = [\arctan(u)]_0^1 = \arctan(1)$$

easy

$$\int_0^{\frac{\pi}{6}} (\sec(x) + \tan(x))^7 \sec(x) \, dx$$

$$\int_0^{\frac{\pi}{6}} (\sec(x) + \tan(x))^7 \sec(x) \, dx = \int_0^{\frac{\pi}{6}} (\sec(x) + \tan(x))^6 \sec(x)(\sec(x) + \tan(x)) \, dx,$$

$$u = \sec(x) + \tan(x), \quad \frac{du}{dx} = \tan(x) \sec(x) + \sec^2(x) = \sec(x)(\sec(x) + \tan(x)),$$

$$dx = \frac{1}{\sec(x)(\sec(x) + \tan(x))} du,$$

$$\int_1^{\sqrt{3}} u^6 \, du = \left[\frac{u^7}{7} \right]_1^{\sqrt{3}} = \frac{\sqrt{3^7}}{7} - \frac{1}{7} = \frac{\sqrt{3^7} - 1}{7}$$

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derivative

easy

medium

integral

beginner

easy